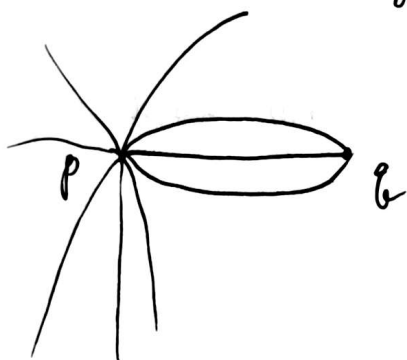


More on Positive Curvature

Def (Conjugate radius)

M is a Riemannian mfd. $p \in M$. The conjugate radius is

$r_{\text{conj}}(p) := \sup \{ r > 0 \mid \text{no conjugate points to } p \text{ exist along any geodesic } \gamma_v(t) \text{ for } \|v\| < r, v \in T_p M \}$.

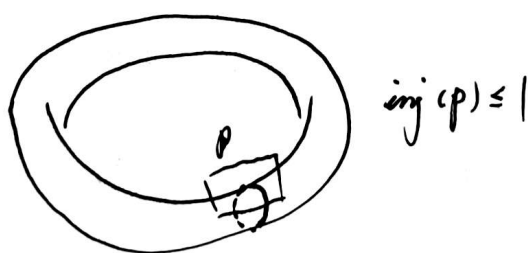
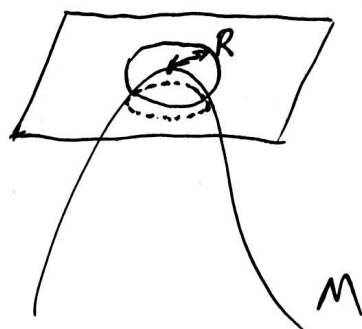
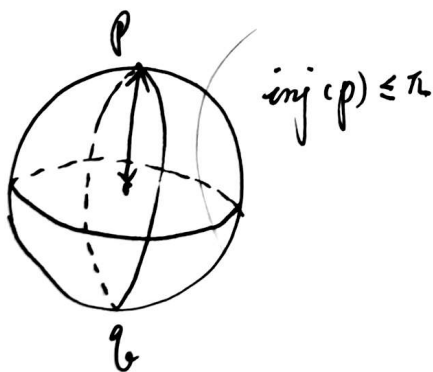


Def (Injectivity radius)

Let M be a Riemannian mfd. Let $\text{inj}_M(p)$ be injectivity radius at point $p \in M$ as the biggest $R > 0$ such that $\exp: B_R(p) \rightarrow M$ is a diffeomorphism.

$$\text{inj}_M := \inf \{ \text{inj}_M(p) : p \in M \}$$

(1)



Thm (Klingenberg, 1959)

If (M, g) is a compact orientable even-dim mfd with $0 < \sec \leq 1$, then, $\text{inj}(M, g) \geq \pi$. If M is not orientable, then $\text{inj}(M, g) \geq \frac{1}{2}\pi$.

Lemma (Klingenberg)

Let (M, g) be a compact Riemannian mfd with $\sec \leq K, K \geq 0$.

Then,

$\text{inj}(p) \geq \min \left\{ \frac{\pi}{\sqrt{K}}, \frac{1}{2} \cdot (\text{length of shortest geodesic loop based at } p) \right\}$.

Proof of Lemma

If $\text{inj}(p) \geq \frac{\pi}{\sqrt{K}}$, we are done.

If $\text{inj}(p) < \frac{\pi}{\sqrt{K}}$, we need $\text{inj}(p) \geq \frac{1}{2}L(p)$, where $L(p)$ is the shortest length of geodesic loop based at p .

Let $r_0 = \text{inj}(p) < \frac{\pi}{\sqrt{K}}$.

As the definition of $\text{inj}(p)$, $\exists q \in M$, s.t. $d(p, q) = r_0$, and a geodesic γ between p and q .

(2)

One of the next case is correct

① q is a conjugate point of p

② $\exists$ another geodesic $\sigma (\neq \gamma)$ of length r_0 between p and q .

According to Rauch Comparison Theorem, since $\sec \leq K$, $K > 0$, $L(p) \geq \frac{\pi}{\sqrt{K}}$. So ① is wrong. q can not be a conjugate point of p . Hence, ② is correct.

Then, $\exists$ two different geodesics γ and σ from p to q . Both of their lengths are r_0 .

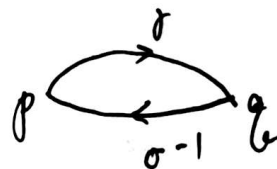
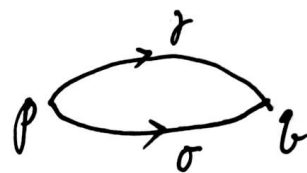
Now, consider the loop $c = \gamma \cup \sigma^{-1}$.

So $\text{length}(c) = 2r_0$. Then $L(p) \leq 2r_0$.

Thus,

$$r_0 \geq \frac{1}{2} L(p).$$

In conclusion, $\text{inj}(p) \geq \min \left\{ \frac{\pi}{\sqrt{K}}, \frac{1}{2} L(p) \right\}$.



Back to the theorem,

Proof:

If M is not orientable, $\pi: \tilde{M} \rightarrow M$ is a covering space, π is a 2-covering map. $\tilde{M}$ is orientable compact mfd.

$$\text{inj}(\tilde{M}) \geq \pi. \quad p \in M, \quad \tilde{p} \in \pi^{-1}(p)$$

$$\text{inj}(p) = \frac{1}{2} \text{inj}(\tilde{p}) \geq \frac{1}{2} \pi$$

③

So we just consider the orientable case.

On the contrary, assume $\text{inj}(M) < \pi$. According to Klingenberg,
 $\exists$ smooth closed geodesic

$$c: [0, 2l] \rightarrow M, \text{ where } l = \text{inj}(M)$$

parametrized arclength. $\text{Length}(c) := L(c) = 2l < 2\pi$.

Because M is even-dim, orientable, $\exists$ a family of closed curves $\{c_\varepsilon\}$, s.t.

$$c_\varepsilon \rightarrow c, (\varepsilon \rightarrow 0), L(c_\varepsilon) < L(c) = 2l.$$

Consider $q_\varepsilon = c_\varepsilon(t_\varepsilon)$, s.t.

$$d(p_\varepsilon, q_\varepsilon) = \max_t d(p_\varepsilon, c_\varepsilon(t))$$

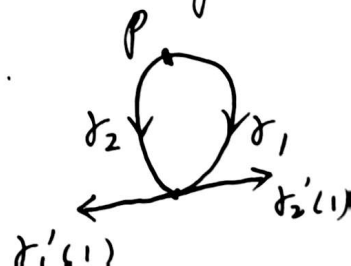
where $p_\varepsilon = c_\varepsilon(0)$.

As $c_\varepsilon \subset B(c_\varepsilon(0), \text{inj}(M))$, then, $\exists!$ segment σ_ε joining p_ε and q_ε ,
 s.t. σ_ε is perpendicular to c_ε at q_ε . As $\varepsilon \rightarrow 0$, $t_\varepsilon \rightarrow \text{inj}(M)$,
 then, σ_ε must subconverge to a segment: σ from $c(0)$ to $c(l)$,
 that is perpendicular c at $c(l)$. (4)

Lemma:

$$p \in M, x = \exp_p(v), \text{ s.t. } \|v\| = \text{inj}(p).$$

If x is not a conjugate point of p . Then, $\exists!$ two geodesics γ_1 and γ_2 between p and x , s.t. $\gamma_1'(1) = -\gamma_2'(1)$.



As $\tau_{\text{inj}}(p) \geq \pi > \text{inj } M = l$, according to the lemma, $\gamma_1(t) = c(t)$, $\gamma_2(t) = c(2l-t)$ are the only two geodesics from $c(0)$ to $c(l)$.

But σ is a third geodesic. Contradiction!

So $\text{inj}(M) \geq \pi$

$$0 < \text{scal} \leq 1, \text{inj}(M, g) \geq \pi$$

□

Def:

$\gamma: [0, a] \rightarrow M^n$ be a geodesic. Denote $V(0, a) = V$ to be the vector space formed by vector field V along γ , which is piecewise differentiable and vanish at the endpoints of γ ($V(0) = V(a) = 0$).

$$I_a(V, W) := \int_0^a \{ \langle V', W' \rangle - \langle R(\gamma', V) \gamma', W \rangle \} dt$$

where $V, W \in V$.

(5)

The index form of γ is

$$Q_a: V \rightarrow \mathbb{R}, Q_a(V) = I_a(V, V)$$

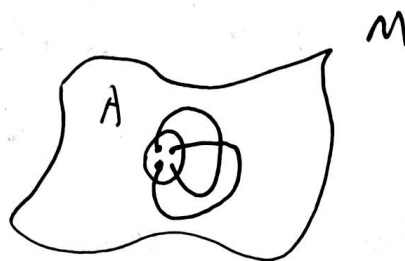
The index of γ is the maximal dim of all subspaces of V on which the quadratic-form associated to I_a is negative definite.

$e_1(t), e_2(t), \dots, e_n(t)$ is a basis of V . The index of

V is the number of negative eigenvalues of the matrix
 $(I_a(e_i(t), e_j(t)))_{i,j}$.

Thm

Let M be a complete Riemannian mfd and $A \subset M$ a compact submfd. If every geodesic in $\perp A, A(M)$ that is perpendicular to A at the end point has index $\geq k$, then $A \subset M$ is k -connected.



Def (Homotopy group)

X is a top space. S^n is an n -dim sphere.

(6)

$$\pi_n(X, x_0) = \{f: (I^n, \partial I^n) \rightarrow (X, x_0)\} / \sim$$

where $I = [0, 1]$ and $\sim$ is homotopy of maps.

$$(I^n, \partial I^n) \xrightarrow{f} (X, x_0) \quad \text{continuous}$$

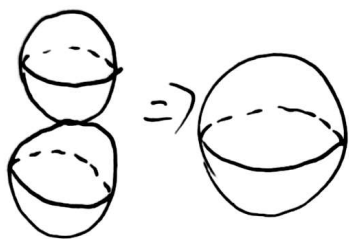
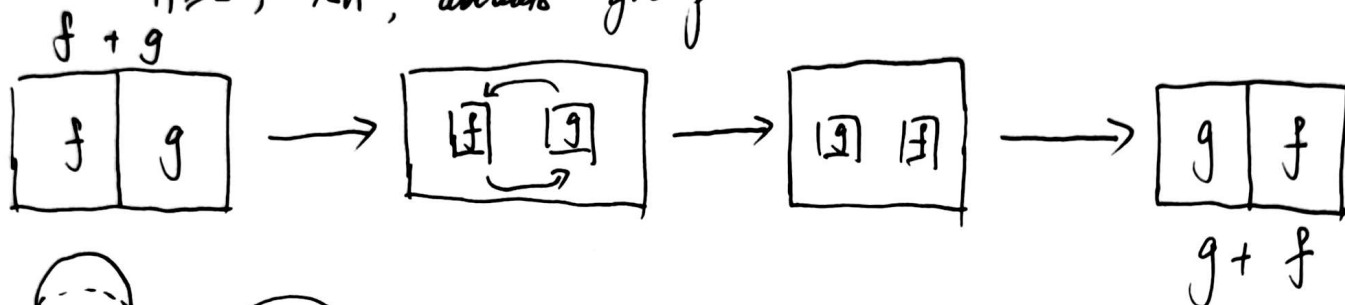
$$\begin{array}{ccc} & & \nearrow \\ (I^n, \partial I^n) & \searrow & (X, x_0) \\ & (I^n / \partial I^n, \partial I^n / \partial I^n) & \\ & \downarrow \cong & \\ & S^n & \downarrow \cong \\ & & \text{pt} \end{array}$$

$$(I^n / \partial I^n, \partial I^n / \partial I^n) \cong (S^n, s_0)$$

$n=0$, $\pi_n \neq \text{group}$

$n=1$, π_1 , fundamental group

$n \geq 2$, π_n , abelian group



(7)